

Dreamcrafter: Immersive Editing of 3D Radiance Fields Through Flexible, Generative Inputs and Outputs

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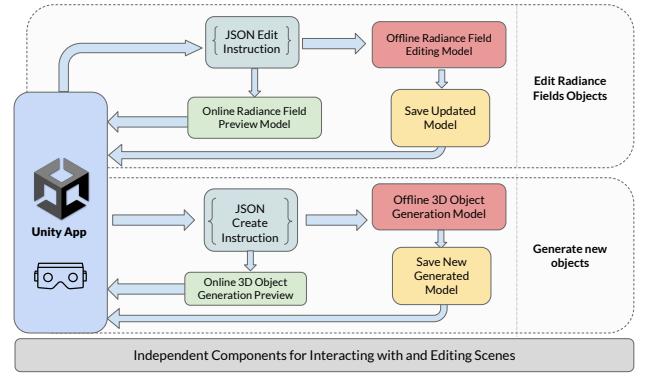


Figure 1: System Overview: (Left) View of edits and spatial annotations made in scene. (Right) Pipeline overview of system including two distinct modules to edit scenes: edit radiance field objects, and create new radiance field objects. Input files generated from Unity are in green, generative model components are in yellow, and saved results are in red.

Abstract

Authoring 3D scenes is a central task for spatial computing applications. Two competing visions for lowering existing barriers are (1) focus on immersive, direct manipulation of 3D content; or (2) leverage AI techniques that capture real scenes (3D Radiance Fields such as NeRFs, 3D Gaussian Splatting) and modify them at a higher level of abstraction, at the cost of high latency. We unify the complementary strengths of these approaches and investigate how to integrate generative AI advances into real-time, immersive 3D Radiance Field editing. We introduce Dreamcrafter, a VR-based 3D scene editing system that: (1) provides a modular architecture to integrate generative AI algorithms; (2) combines different levels

of control for creating objects, including natural language and direct manipulation; and (3) introduces proxy representations that support interaction during high-latency operations.

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1 Introduction

Spatial computing applications such as Augmented and Virtual Reality rely on 3D content and scenes. Thus, creating appropriate tools for authoring and editing 3D content has been a long-standing key challenge for HCI researchers.

Traditionally, mesh-and-texture-based approaches have been used to author 3D content. Various research efforts to introduce better editing techniques notwithstanding (e.g., [1, 3]), the expertise hurdle to create and modify 3D content in this way has been high,

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generally leaving such authoring to a small number of expert users. Immersive systems like Virtual Dream Reliving [6] and Calliope [9] demonstrate using AI as a collaborator in scene generation.

More recently, two additional developments hold the promise of reducing authoring burdens. First, novel approaches for representing 3D scenes based on radiance fields (e.g., NeRFs [7] and 3D Gaussian Splatting [5]) allow for straightforward capture of photorealistic environments from real scenes using common cameras, instead of having to model objects from scratch. Second, generative AI developments have introduced novel ways of editing radiance field scenes at higher levels of abstraction, e.g. through text instructions (as in Instruct-NeRF2NeRF [2]). While offering the ability to edit at a semantic level rather than a lower geometry level, such techniques also tend to be compute-intensive and not yet amenable to run in realtime.

We introduce Dreamcrafter, a Virtual Reality 3D content generation and editing system assisted by generative AI. The core idea behind Dreamcrafter is to use direct manipulation for spatial positioning and layout; and leverage generative AI for editing style and appearance of objects. Because generative AI edits are unlikely to run in real-time, Dreamcrafter introduces rapid proxy representations, e.g. using a 2D diffusion model to create a stand-in image for a longer-running 3D generative task.

2 System Design and Implementation

Dreamcrafter provides an interface to edit and generate radiance field objects (3DGS in this case) using modular generative AI-enabled tools. Dreamcrafter supports different levels of user control and gives real-time proxy representations to preview time-consuming edits and introduces new workflows leveraging image diffusion models (i.e., Stable Diffusion). Dreamcrafter makes Gradio API calls through a flask server from Unity.

2.1 Key interactions

Dreamcrafter supports four interactions for moving, editing, and generating new radiance field objects.

(1) Move objects: Users can position, rotate, or scale objects with spatial manipulations with hand movements and VR controls. Objects can be positioned, rotated, or scaled within the scene.

(2) Edit radiance field objects via prompting: Radiance field objects can be given stylistic or basic structural edits by pointing at an object and speaking an instruction, e.g. “Make this chair chrome and futuristic.”. A render of the object is given to the Instruct-Pix2Pix module, which applies the instruction to show as a 2D preview of the edit. Users can select from three edit variants.

(3) Generate objects via prompting: Users can generate objects by pointing at the ground and speaking a prompt of the object they want to create (Figure 2). This sends an API call to the 3D generative module, Shap-E [4], which generates a low fidelity mesh and NeRF from a text prompt. A render of the generated NeRF is stylized using depth conditioned ControlNet [11] with the initial prompt. The user can select from three stylized 2D image variants of the object. During an offline process, the full fidelity 3D objects are generated and placed in the scene.

(4) Sculpt then stylize objects: Alternatively, users can generate objects by creating an arrangement of basic 3D primitives (i.e., spheres, cubes, and cylinders) (Figure 2). Users then take a snapshot

of this arrangement, which is stylized with ControlNet based on a user-given prompt. Once the user confirms the stylized and sculpted generation, the object can be placed in the scene.

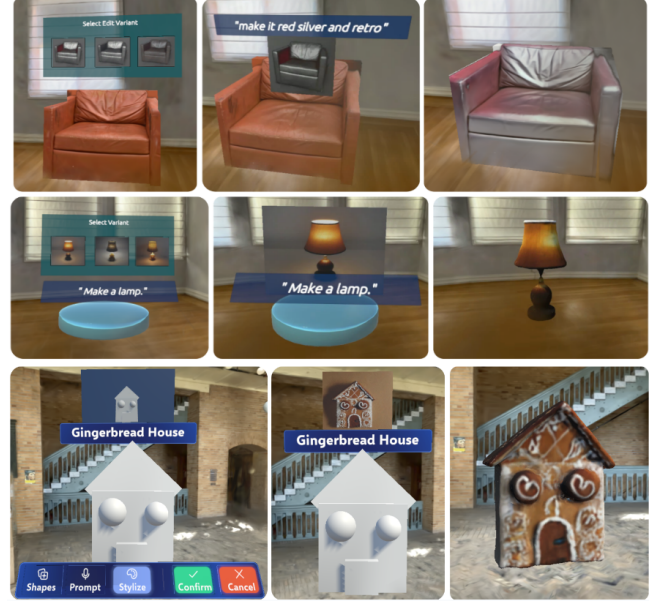


Figure 2: Edit and Generate Interactions (Top) Radiance Field Object Editing with preview (Center) Object Generation via Prompting (Bottom) Object Generation via Sculpting

2.2 Proxy representations: Labels and Previews

Proxy representations are intended to help users see the impact of their editing operations in real time. There are two types of proxy representations: labels and image previews. Figure 2 (center) and Figure 2 (center) show the labels and image previews. The labels show the prompts users have spoken aloud as commands to the generative AI modules (e.g., “make the sofa blue”). The image previews show 2D versions of the anticipated generation. These image previews are generated using Instruct-Pix2Pix which is the underlying 2D image editing system used for the 3D radiance field editing system, Instruct-NeRF2NeRF.

Both the labels and image previews are associated with radiance field objects in the scene. This is done through a spatial annotation framework we developed. The framework logs each object’s positions, object type, generative AI prompt, and image preview to a JSON file used for 3D generation and replacement.

2.3 Generative modules and Offline Processing

Using the JSON log output from the spatial annotation system, Dreamcrafter makes instruction and tool specific API calls for each generative AI module. A Python broker server receives a server message from the Unity project and forwards instruction parameters (e.g., instruction type, text prompt, image input) to the specified module. Figure 1 shows an overview of the system architecture. Object generation uses a 3D generative module Shap-E, and a 2D image stylization module ControlNet and Stable Diffusion. The full object 3D generations use 2D-image to 3D-model models such as LGM [8] or a text-to-3D system. The final 3D object edits are

done using Instruct-GS2GS [10] for Gaussian Splatting objects. The modules are exchangeable and can be implemented to use newer AI models. After the final objects are added to the scene, users can re-edit the scene again, creating an iterative design process.

3 Conclusion

Dreamcrafter uses direct manipulation for spatial positioning and layout and leverages generative AI for editing style and appearance of objects. Because AI edits are unlikely to run in real-time, Dreamcrafter introduces rapid proxy representations, e.g. using a 2D diffusion model to create a stand-in image for a longer-running 3D generative task.

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